

Contextual and Bayesian bandits

Lecture 2 — Contextual Bandits. Book chapter 3.

What this note uses from earlier lectures.

- Lecture 1 — Bandits & Regret, slides 1–63, in `L01-1-bandits.pdf`: Theorem 5, Theorem 8, Corollary 9

Definition 1. Contextual bandit

A contextual bandit is a distribution ρ over a context set $\mathcal{Z}$ together with a mean-reward function $r : \mathcal{Z} \times \mathcal{A} \rightarrow [0, 1]$, where $|\mathcal{A}| = K$. For $t = 0, 1, \dots, T - 1$:

1. a context $z_t \sim \rho$ is drawn i.i.d. and revealed;
2. the learner picks $A_t \in \mathcal{A}$, possibly at random, using z_t and the history;
3. a reward r_t with $\mathbb{E}[r_t \mid z_t, A_t] = r(z_t, A_t)$ is revealed, and nothing is revealed about the other $K - 1$ actions.

A common simplification takes $r_t = r(z_t, A_t)$ deterministically, on the grounds that the context, not the noise, is the new difficulty. Every result below holds either way.

Definition 2. Contextual regret

$$\text{Reg}(T) = TV(\pi^*) - \sum_{t=0}^{T-1} V(\pi_t),$$

where π_t is the policy the learner is using at round t .

Definition 3. Inverse propensity scoring

Suppose at round t the action was drawn from a known distribution p_t with $p_t(a) > 0$ for all a . The IPS reward vector is

$$\hat{\mathbf{r}}_t[a] = \frac{r_t \mathbf{1}[A_t = a]}{p_t(a)}, \quad a \in \mathcal{A}.$$

Under uniform exploration, $p_t(a) = 1/K$ and this is Kr_t on the action played and 0 everywhere else.

Lemma 4. IPS is unbiased

For every context z_t and every action a , $\mathbb{E}_{A_t \sim p_t}[\hat{\mathbf{r}}_t[a] \mid z_t] = r(z_t, a)$, and consequently $\mathbb{E}[\hat{\mathbf{r}}_t[\pi(z_t)]] = V(\pi)$ for every π .

Proof.

Condition on z_t and take the expectation over $A_t \sim p_t$ and the reward noise:

$$\begin{aligned} \mathbb{E} \left[\frac{r_t \mathbf{1}[A_t = a]}{p_t(a)} \right] &= \sum_{a' \in \mathcal{A}} p_t(a') \frac{\mathbb{E}[r_t \mid z_t, a'] \mathbf{1}[a' = a]}{p_t(a)} \\ &= p_t(a) \frac{r(z_t, a)}{p_t(a)} = r(z_t, a). \end{aligned}$$

Averaging over $z_t \sim \rho$ gives $\mathbb{E}[\hat{\mathbf{r}}_t[\pi(z_t)]] = V(\pi)$. The division by $p_t(a)$ is what removes the asymmetry the sampling introduced: an action played half as often has its rewards counted twice as heavily.

Lemma 5. What unbiasedness costs

Under uniform exploration, $0 \leq \hat{\mathbf{r}}_t[a] \leq K$ and $\text{Var}(\hat{\mathbf{r}}_t[a] \mid z_t) \leq K$.

Proof.

The estimator is $Kr_t \leq K$ on one action and 0 elsewhere, giving the range. For the variance, $\mathbb{E}[\hat{\mathbf{r}}_t[a]^2 \mid z_t] = \frac{1}{K} (K)^2 \mathbb{E}[r_t^2 \mid z_t, a] \leq K$, since rewards lie in $[0, 1]$, and the variance is at most the second moment. **Algorithm A — Explore-then-commit for contextual bandits**

Input: horizon T , policy class Π , exploration budget N .

1. **Explore.** For $t = 0, \dots, N-1$: observe z_t , draw $A_t \sim \text{Unif}(\mathcal{A})$, observe r_t , and record the IPS vector $\hat{\mathbf{r}}_t$.
2. **Fit.** $\hat{\pi} \in \arg \max_{\pi \in \Pi} \sum_{t=0}^{N-1} \hat{\mathbf{r}}_t[\pi(z_t)]$. 3. **Commit.** For $t = N, \dots, T-1$: observe z_t , play $\hat{\pi}(z_t)$.

Theorem 6. Regret of contextual explore-then-commit

Let Π be finite. Run Algorithm A with $1 \leq N \leq T$. With probability at least $1 - \delta$,

$$\text{Reg}(T) \leq N + TK \sqrt{\frac{2 \ln(2|\Pi|/\delta)}{N}},$$

and choosing N to minimise the right-hand side gives

$$\text{Reg}(T) \leq 3 \cdot 2^{-1/3} T^{2/3} K^{2/3} \ln^{1/3} \left(\frac{2|\Pi|}{\delta} \right).$$

Proof.

Write $\hat{V}(\pi) = \frac{1}{N} \sum_{t=0}^{N-1} \hat{\mathbf{r}}_t[\pi(z_t)]$. By Lemma 4 the summands are i.i.d. with mean $V(\pi)$, and by Lemma 5 they lie in $[0, K]$. Hoeffding's inequality (Theorem 5 (L01-1), rescaled by K) gives, for one fixed π and any $\delta' \in (0, 1)$,

$$\mathbb{P} \left(|\hat{V}(\pi) - V(\pi)| > K \sqrt{\frac{\ln(2/\delta')}{2N}} \right) \leq \delta'.$$

Take $\delta' = \delta/|\Pi|$ and a union bound over the $|\Pi|$ policies: with probability at least $1 - \delta$, every $\pi \in \Pi$ satisfies $|\hat{V}(\pi) - V(\pi)| \leq \varepsilon$ with $\varepsilon = K \sqrt{\ln(2|\Pi|/\delta)/(2N)}$.

On that event, the same three-step chain as in Theorem 8 (L01-1) applies to $\hat{\pi}$, which maximises $\hat{V}$:

$$V(\pi^*) \leq \hat{V}(\pi^*) + \varepsilon \leq \hat{V}(\hat{\pi}) + \varepsilon \leq V(\hat{\pi}) + 2\varepsilon.$$

Exploration contributes at most N , since each round's regret is at most 1; commitment contributes at most $(T - N) \cdot 2\varepsilon \leq 2T\varepsilon$. Adding them gives the first display.

For the second, set $c = \sqrt{2 \ln(2|\Pi|/\delta)}$, so the bound reads $f(N) = N + TKcN^{-1/2}$. As in Corollary 9 (L01-1), f is convex on $N > 0$ and $f'(N) = 1 - \frac{1}{2}TKcN^{-3/2}$ vanishes at $N^* = (TKc/2)^{2/3}$; writing $u = (TKc/2)^{1/3}$ gives $f(N^*) = u^2 + 2u^2 = 3(TKc/2)^{2/3}$, and substituting $c^{2/3} = 2^{1/3} \ln^{1/3}(2|\Pi|/\delta)$ gives the claim. **Algorithm B — ϵ -greedy for contextual bandits**

For $t = 0, 1, \dots, T - 1$: observe z_t ; draw

$$A_t \sim p_t = (1 - \epsilon_t) \delta(\pi^t(z_t)) + \epsilon_t \text{Unif}(\mathcal{A}),$$

observe r_t , form $\hat{\mathbf{r}}_t$ with the propensity $p_t(A_t)$ actually used, and refit $\pi^{t+1} \in \arg \max_{\pi \in \Pi} \sum_{i \leq t} \hat{\mathbf{r}}_i[\pi(z_i)]$. Decay ϵ_t towards zero.

Proposition 7. Gaussian conjugacy

Let the unknown mean of an arm have prior $\mu \sim \mathcal{N}(m_0, s_0^2)$, and let $r_1, \dots, r_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Then the posterior is $\mathcal{N}(m_n, s_n^2)$ with

$$\frac{1}{s_n^2} = \frac{1}{s_0^2} + \frac{n}{\sigma^2}, \quad m_n = s_n^2 \left(\frac{m_0}{s_0^2} + \frac{1}{\sigma^2} \sum_{i=1}^n r_i \right).$$

Proof.

By Bayes' rule the posterior density is proportional to prior times likelihood:

$$p(\mu \mid r_{1:n}) \propto \exp\left(-\frac{(\mu - m_0)^2}{2s_0^2}\right) \prod_{i=1}^n \exp\left(-\frac{(r_i - \mu)^2}{2\sigma^2}\right).$$

The exponent is a quadratic in μ ; collecting terms,

$$-\frac{1}{2} \left[\mu^2 \left(\frac{1}{s_0^2} + \frac{n}{\sigma^2} \right) - 2\mu \left(\frac{m_0}{s_0^2} + \frac{\sum_i r_i}{\sigma^2} \right) \right] + \text{const},$$

which is $-\frac{1}{2s_n^2}(\mu - m_n)^2 + \text{const}$ for exactly the s_n^2 and m_n in the statement. A density proportional to a Gaussian kernel is that Gaussian.

Proposition 8. Beta-Bernoulli conjugacy

If $\mu \sim \text{Beta}(\alpha, \beta)$ and $r \sim \text{Bernoulli}(\mu)$, then $\mu \mid r \sim \text{Beta}(\alpha + r, \beta + 1 - r)$. After k successes in n trials from the uniform prior $\text{Beta}(1, 1)$, the posterior is $\text{Beta}(1 + k, 1 + n - k)$, with mean $\frac{k+1}{n+2}$ and variance $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Proof.

The Beta density is proportional to $\mu^{\alpha-1}(1-\mu)^{\beta-1}$ and the Bernoulli likelihood is $\mu^r(1-\mu)^{1-r}$. Their product is proportional to $\mu^{\alpha+r-1}(1-\mu)^{\beta+(1-r)-1}$, the kernel of $\text{Beta}(\alpha+r, \beta+1-r)$. Iterating n times adds the successes to α and the failures to β ; the moments are those of the Beta distribution.

Algorithm C — Thompson sampling (posterior sampling)

For $t = 0, 1, \dots, T - 1$:

1. for each arm a , draw $\tilde{\mu}_t(a)$ independently from the current posterior of $\mu(a)$;
2. play $A_t \in \arg \max_a \tilde{\mu}_t(a)$;
3. observe r_t and update the posterior of arm A_t only.

In the contextual case the posterior is over the parameters of the reward model, and step 1 draws one model; step 2 acts greedily *with respect to that draw*.

Proposition 9. Thompson sampling is probability matching

Assume the arms' means are independent under the posterior given the history h_t . Then for every arm a ,

$$\mathbb{P}(A_t = a \mid h_t) = \mathbb{P}(a \in \arg \max_{a'} \mu(a') \mid h_t)$$

(ties broken by the same rule on both sides). The algorithm plays each arm exactly as often as it believes that arm is optimal.

Proof.

By independence, drawing $\tilde{\mu}_t(a)$ from each arm's marginal posterior independently is a draw of the whole vector $\tilde{\mu}_t$ from the joint posterior of $(\mu(1), \dots, \mu(K))$ given h_t . The event $\{A_t = a\}$ is the event $\{a \in \arg \max_{a'} \tilde{\mu}_t(a')\}$, whose probability under the joint posterior is by definition the posterior probability that a is an optimal arm.